PERSPECTIVES IN CONTINUOUS BIOPROCESSING

INVENTING FOR LIFE

BioNJ Manufacturing Briefing
21 September 2018

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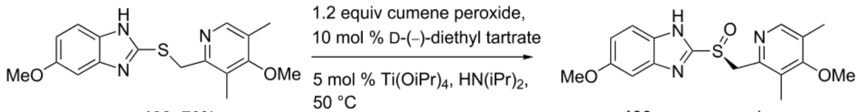
Biologics vs. Small Molecules

Small Molecules→ Chemical Synthesis

Drug Substance (API)
- Solid
- Storage @ room temp
- Unformulated

Drug Product
- Tablets
- Room temperature storage
- Oral dosing



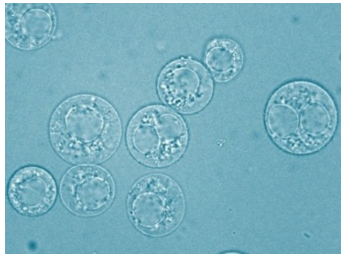
199, 70%

190, esomeprazole, 74%, >99.9% ee

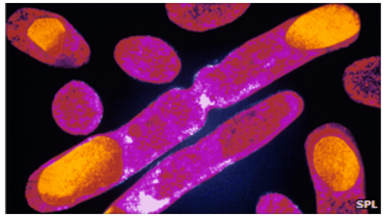
Biologics→ Fermentation

Drug Substance
- Liquid
- Storage @ -70C
- Formulated
  - Buffer, sugar, surfactant

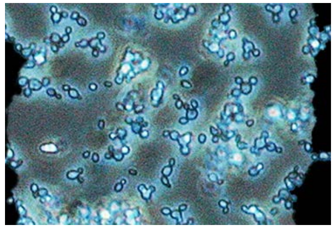
Drug Product
- Liquid or lyophilized
- Storage @ 5C
- Intravenous or subcutaneous dosing



CHO Cells

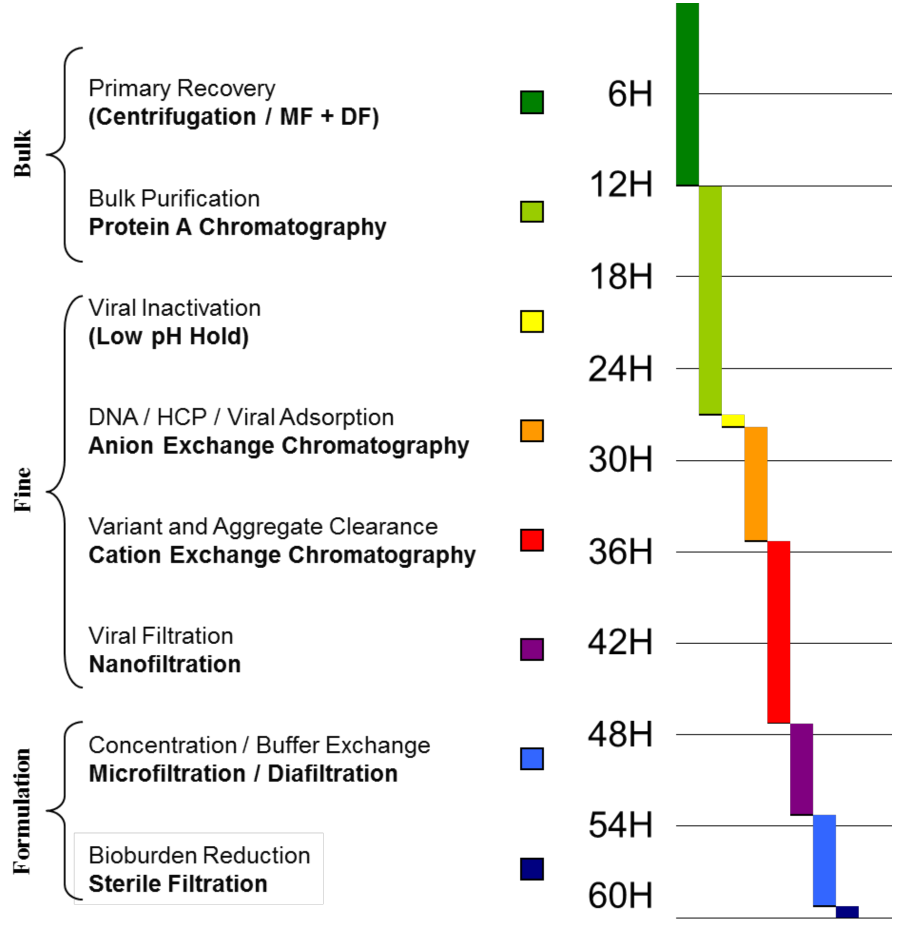


E. coli

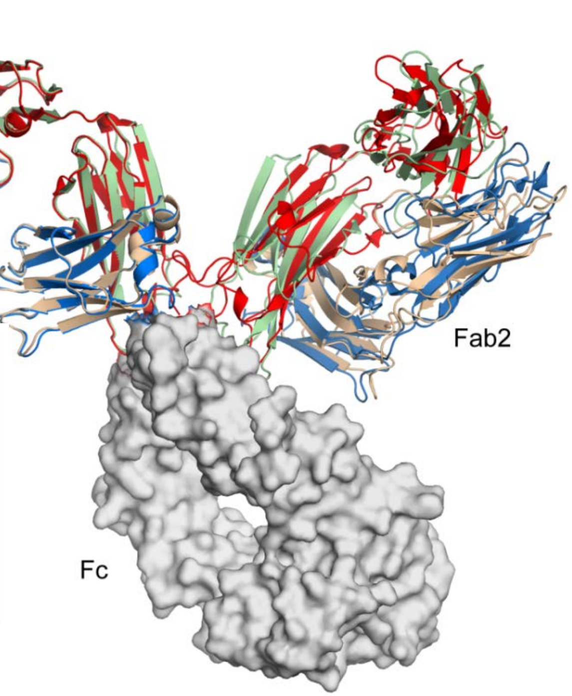


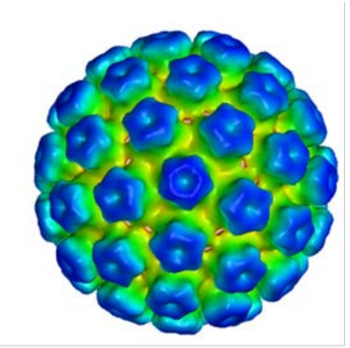
Pichia

Typical Biologics Purification



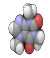
Beyond Antibodies



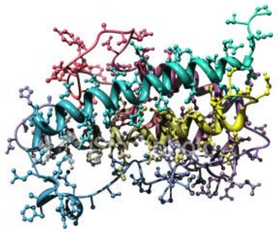


GARDASIL 9
Human Papillomavirus
9-valent Vaccine, Recombinant

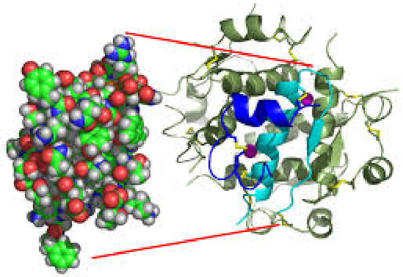
Growth Factors & Cytokines
~ 15,000 -25,000 Da
Example: INTRON A



“Small Molecule”
194 Da
Example: Januvia

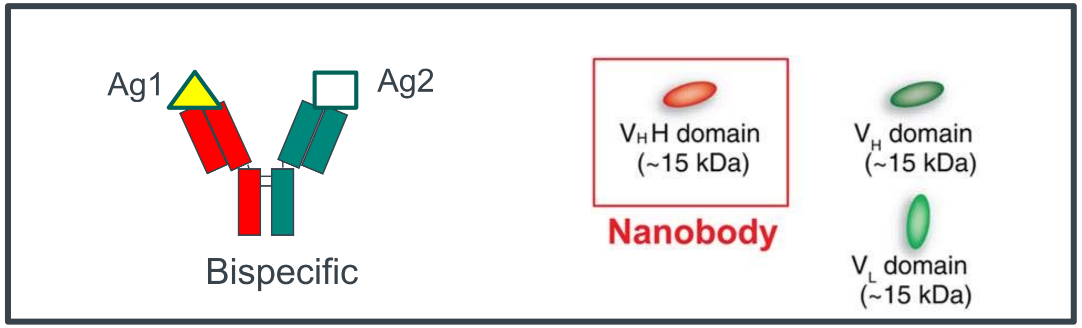


Peptides- MW ~22,000 Da



Insulin- MW ~5800 Da

Monoclonal Antibody (mAb)
~150,000 Da



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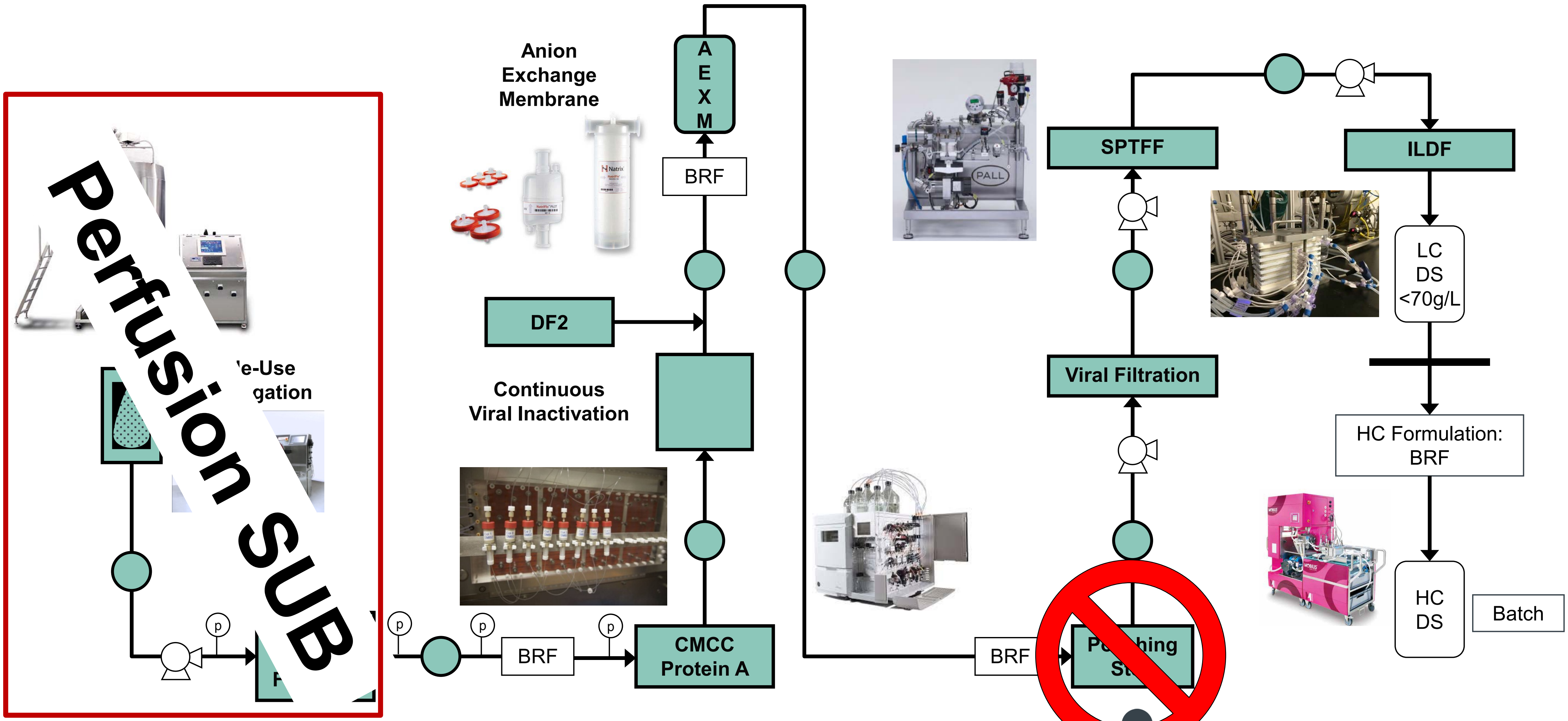
Continuous Processing Vision - 2,000L SUB*



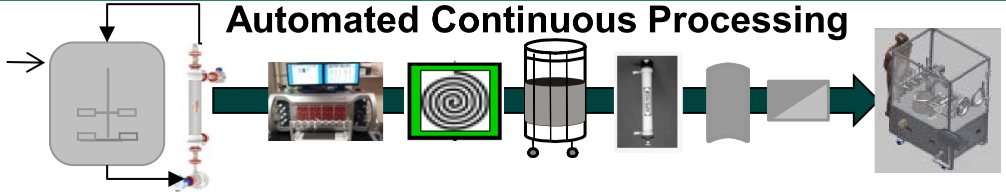
*Single-Use Bioreactor

Single-Use Surge Vessel

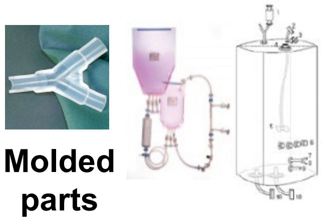
Overall DSP Time Cycle is Dictated by the Longest Step
Other Steps are Lengthened to Compensate

Vision for Biologics Agile Supply

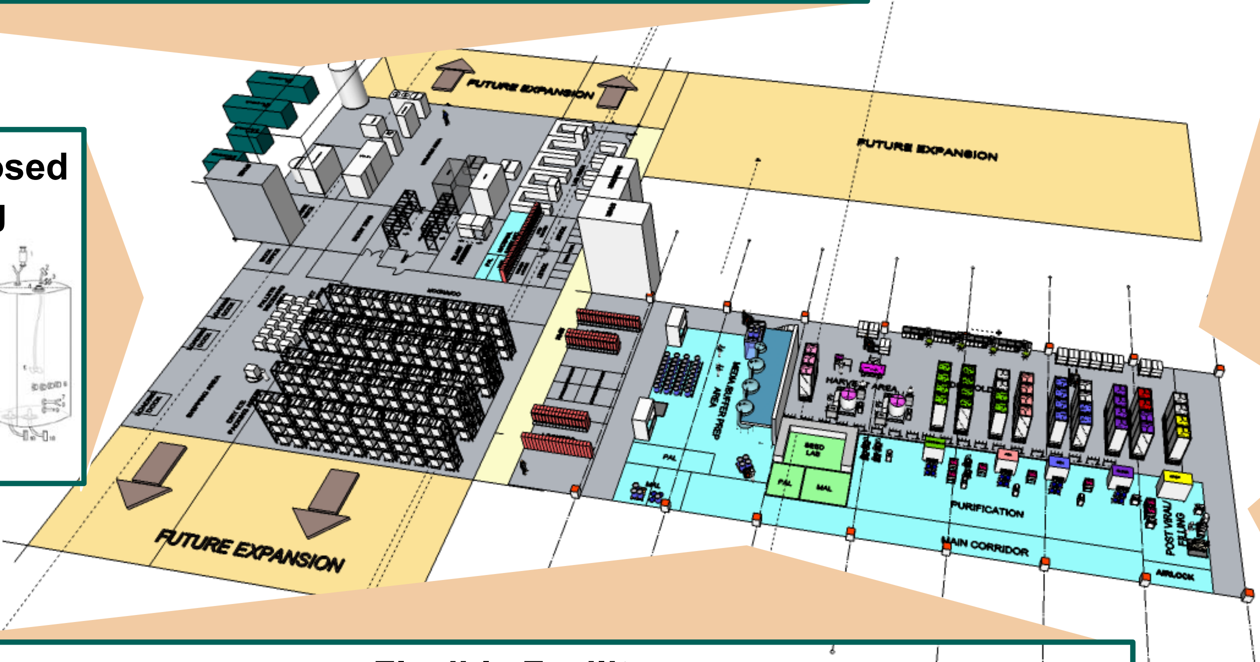
Automated Continuous Processing



Single Use Closed Processing

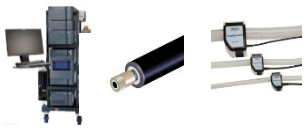


Molded parts

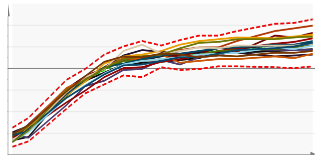


Adaptive Process Control

PAT tools

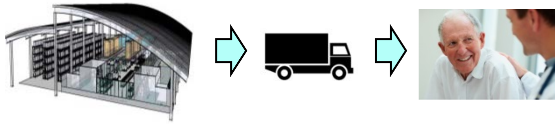


Predictive MDVA models



Real Time Release Testing
QC Shop floor
Consistent Quality assurance

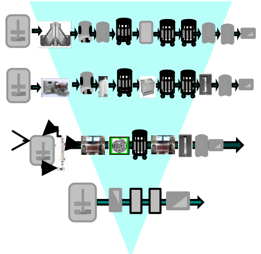
Synchronized supply chain



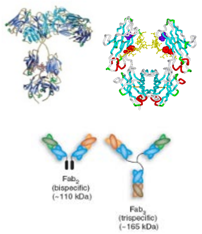
On demand production
Portable capacity for
Local or Regional Options

Flexible Facility

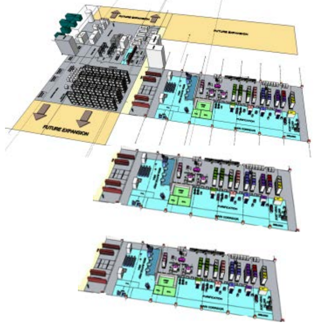
Plug & play
Toolbox



Multi
Product



Expansion
via Scale out



v1.0

v1.2

Innovation
continuum

v2.0

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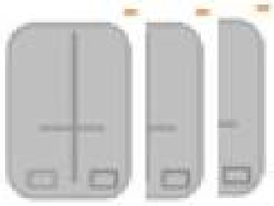
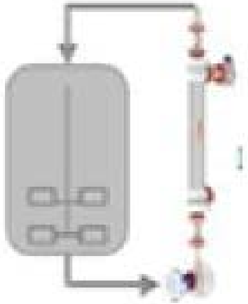
Perfusion Cell Culture

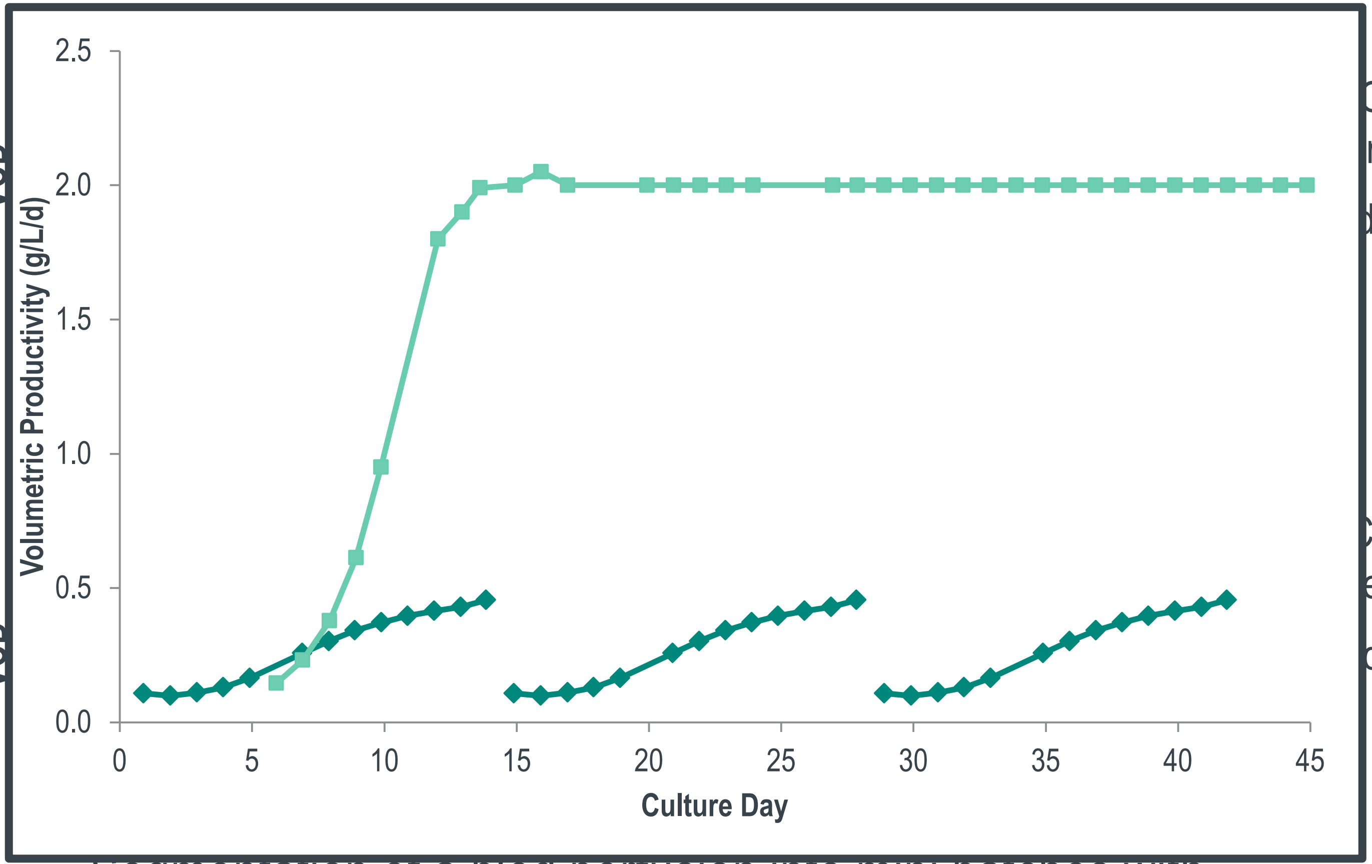
Quality vs. time: Mini-batch concept

Multi
Fed-Batch



Bled
Perfusion





QA
rage
deviation

CQA
erage
deviation

Segmentation of a bled perfusion into mini batches with predictable product quality enables real-time release

*Curtesy of Nuno Pinto

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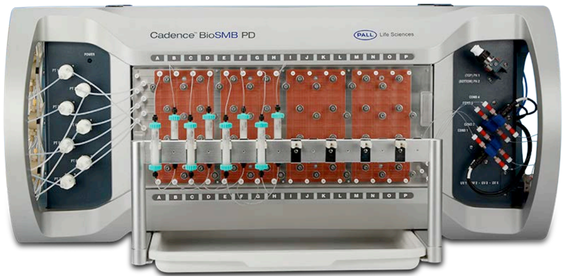
Continuous Chromatography (mAb 1)

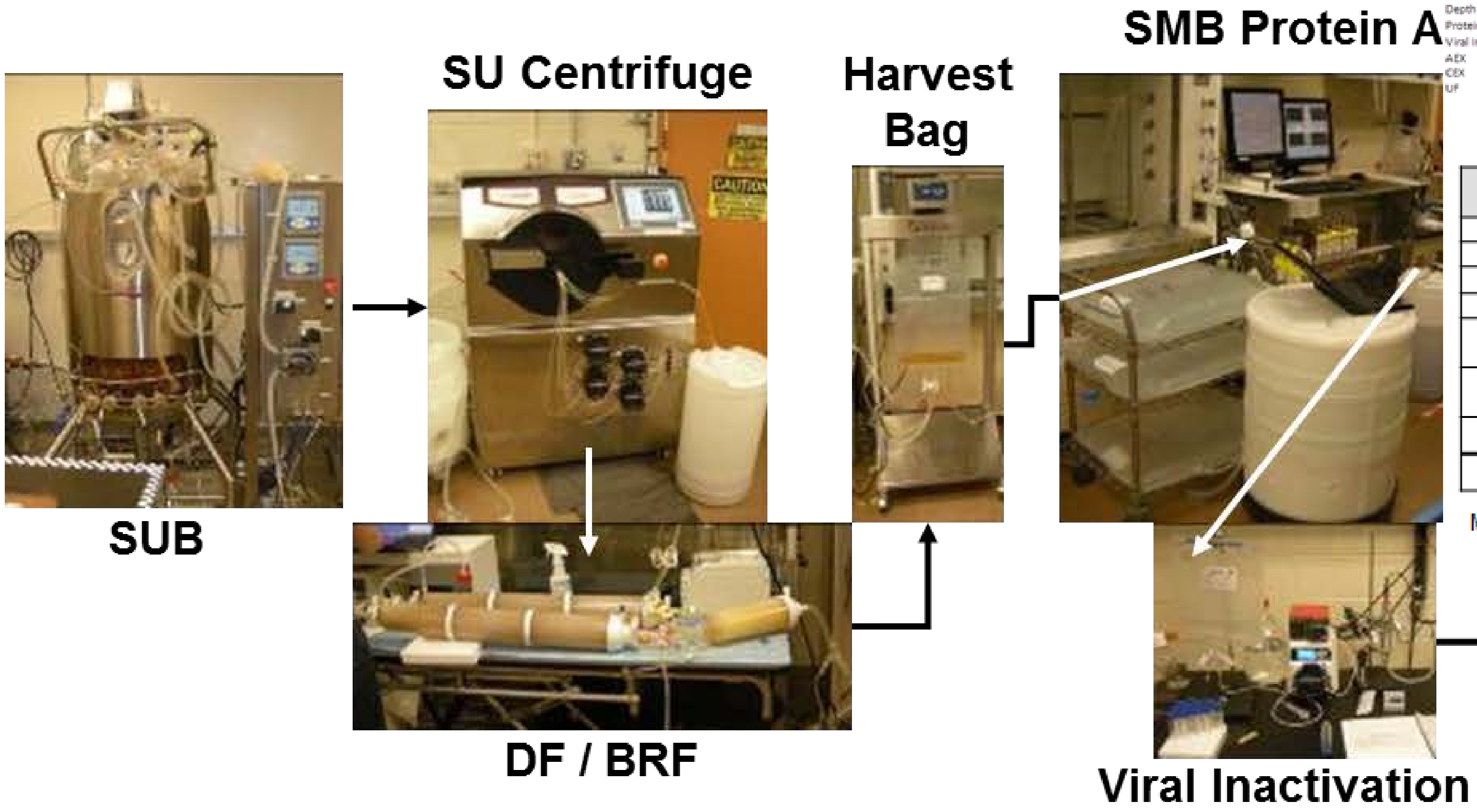
Process Comparison Across Scales & Modes



	Batch Reference	Cadence BioSMB PD	Cadence BioSMB Process
Product Concentration	9.54 g/L	13.84 g/L	13.85 g/L
Product yield	98%	97%	97%
Aggregate	0.45%	0.72%	0.65%
LRV DNA	n.a.	4.16	5.02
LRV HCP	2.4	2.6	2.5
Specific Productivity	16 g/(L·h)	56 g/(L·h)	56 g/(L·h)
Comparable Process			
Cycles	3	[6]	13
Column Diameter	40 cm	1.13 cm	14 cm
Column Height	16 cm	5 cm	5 cm
Adsorbent Volume	20 L	0.025 L	3.85 L
Flow rate	315 L/h	0.24 L/h	37 L/h

- Feed material: 400 L clarified cell supernatant with 5.8 g/L mAb
- Chromatography sorbent: KANEKA KanCapA Protein A media
- **→ 2.3 Kg from 400 L processed in 13 cycles over 11 hours**

Continuous Processing Case Study (mAb 2)



Continuous Processing Case Study mAb 2
- Platform

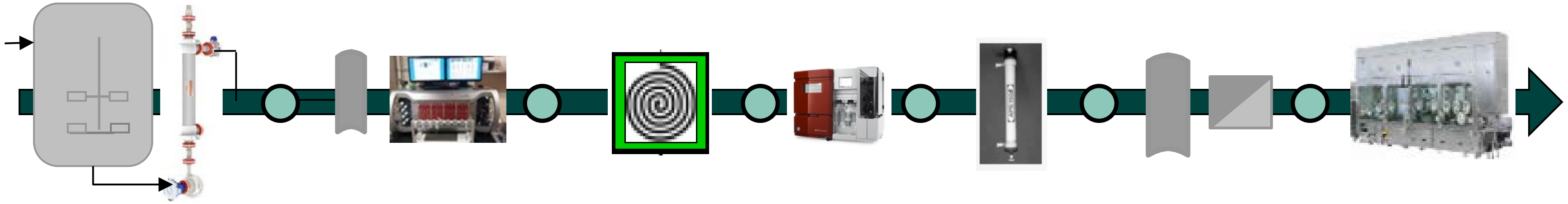
	Average Yield	DNA* [ppm]	HCP [ppm]	Res. ProA [ppm]	% Monomer
Centrifugation	97.3%	N/S	N/S	N/S	N/S
DF/BRF	98.6%	30,515	383,300	N/S	N/S
Protein A SMB	98.1%	N/S	N/S	N/S	N/S
Viral Inactivation	100%	2	1,063	2.1	89.8%
Anion Exchange Membrane	98.8%	<LOQ	82	1.5	99.0%
Cation Exchange Chromatography	84.2%	<LOQ	605	<LOQ	99.2%
SPTFF	99.5%	0.001	35	<LOQ	99.0%
Overall	77.9%	0.001	8.7	<LOQ	99.0%

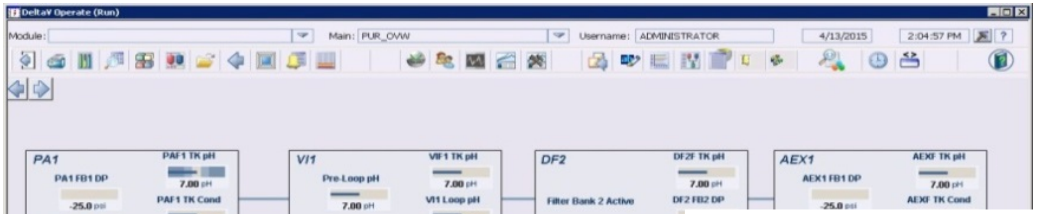
Mass Balance = 93%

Cation Exchange

Automated mAb production:

Protein Refinery Operations Lab (PROLab)





- Single Use Closed Processing
- Overarching Delta-V Control
- Integrated PAT / SIPAT Connectivity

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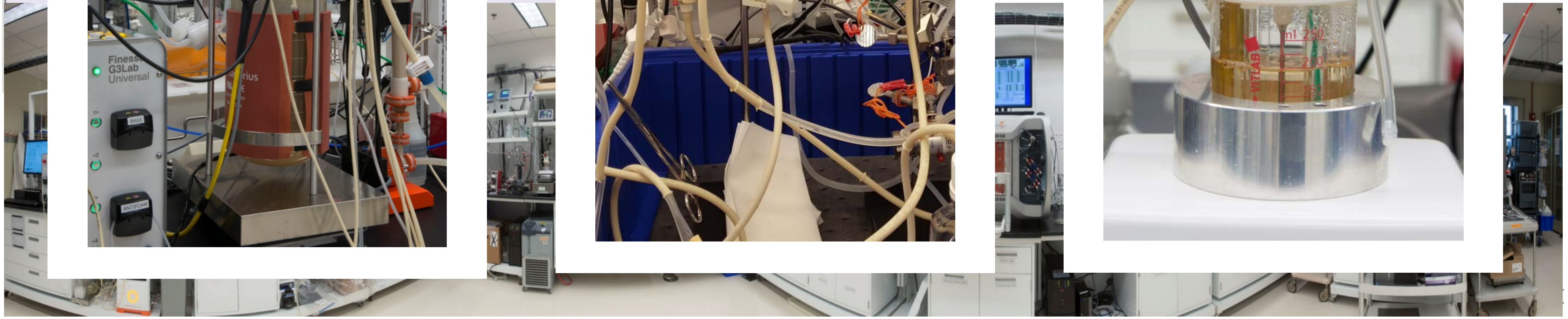
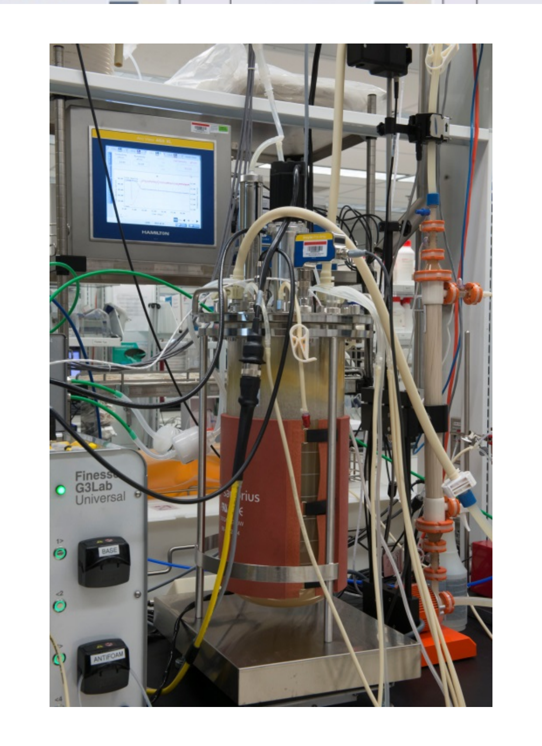
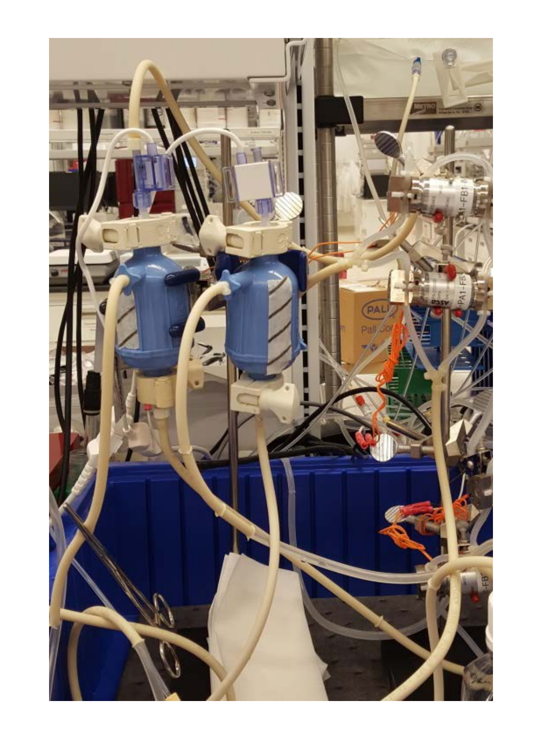
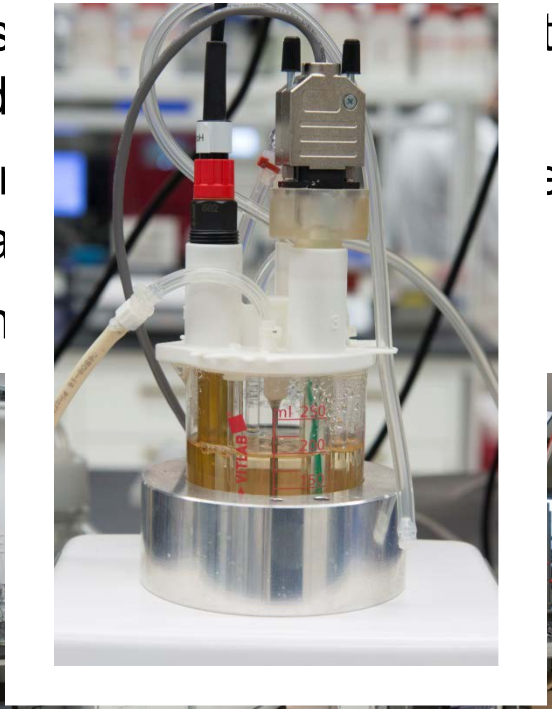


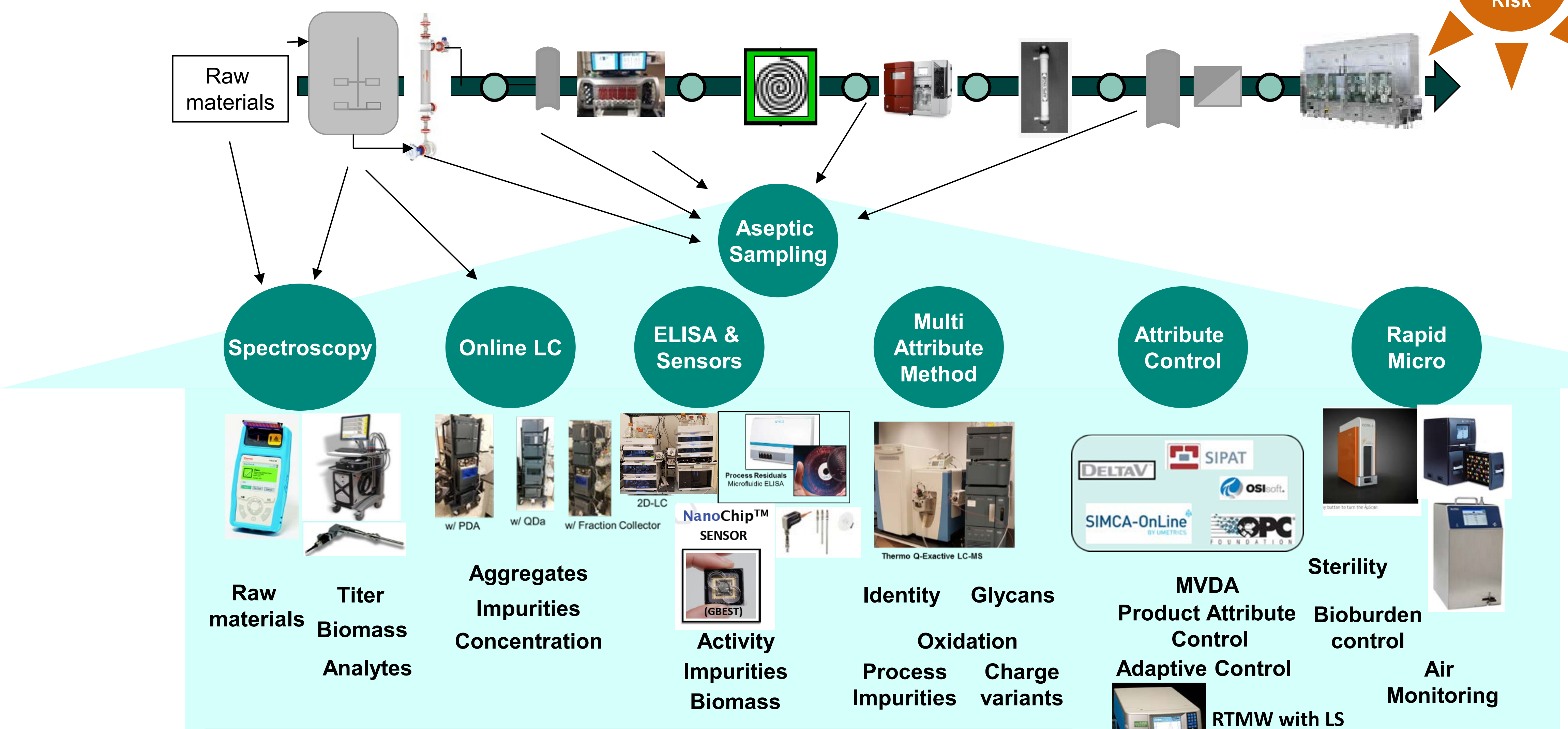






Enabling Technologies for Real Time Analytics:

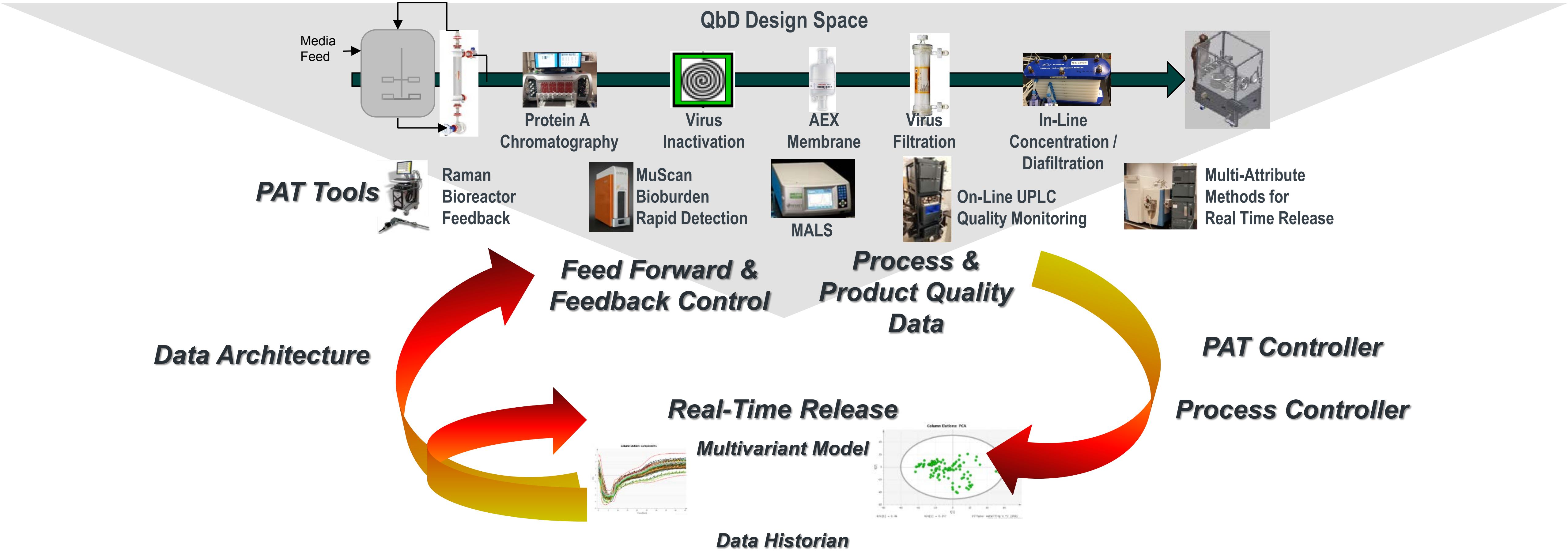
Analytical QC transition to the shop floor



- Developed for integration into continuous processing
- Deployable into current batch processes

Process, PAT, and IT Toolbox

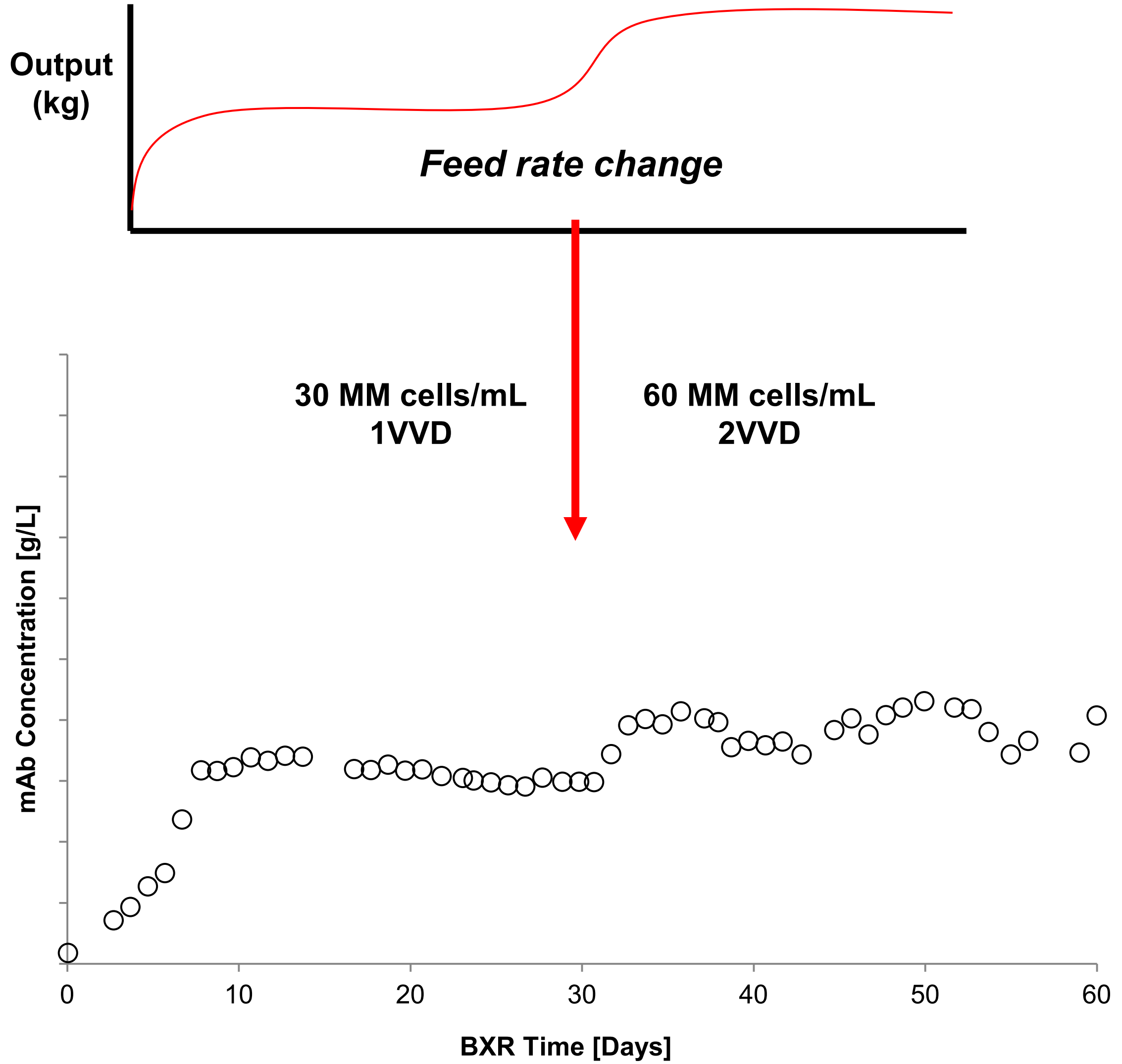
Connecting systems for robust operations

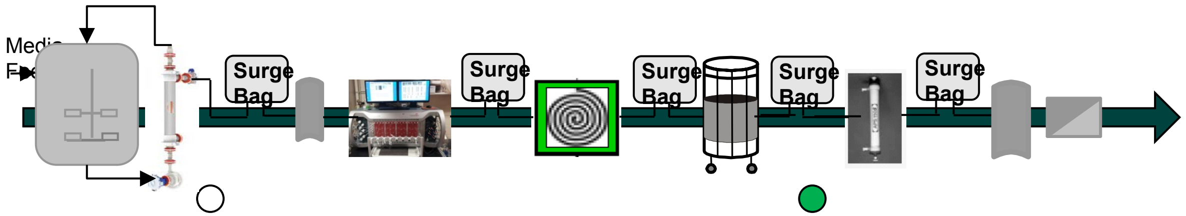


Continuous Downstream Operation

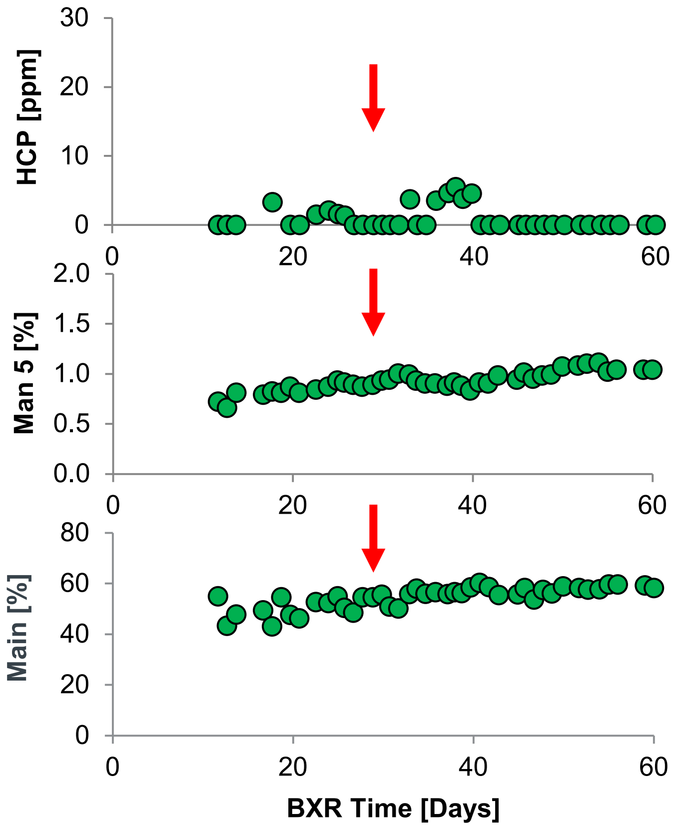
Process control for flexible drug supply

Perfusion feed rate increase to double kg output





Minimal impact to purity & quality



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Process Validation for CM in Biologics-Under Control

Start-up and Shut down

Run time Evaluation

Process Excursion and Response

Change Impact

Established principles, guides, standards, quantitative statistical methods for traditional batch manufacturing applies to continuous manufacturing

#vials thawed

Continuous process verification (per ICH Q8) as an alternative validation approach can use in-line, on-line, or at-line monitoring or controls to evaluate process performance

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*Curtesy of Lucy Chang: DIA 2018 Global Annual Meeting

Technical Challenges for Continuous Biologics (DSP)

- Facility design considerations
- Residence time distribution modeling
- Microbiology and Adventitious Agents control strategy
- Scale down models & change of PD workflows
- Viral safety validation
- Robust aseptic sampling and clarification technologies
- Demonstration of robust process scale GMP systems
- Robust single use sensors and consumables and supply
- Connectivity
    - Plug and Play automation solutions
    - Automated data movement and process control
    - Automation hurdles for lab scale systems

<u>Process Integration</u>

- Data standardization
- Linkage of discrete and continuous data
- Moving past batch mentality
- Robust and agile quality systems
- Electronic everything
- Development PAT to Manufacturing

Acknowledgements

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Lucy Chang

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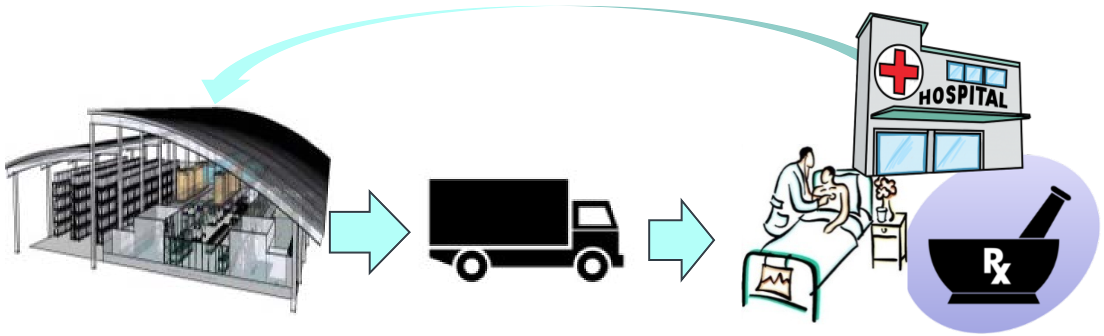
Maury Bayer (Proconex)

Summary and Future Direction

- Merck PROLab designed as a fully automated continuous production suite for testing for next generation bioprocesses
- Continued development on a toolbox of processing platforms
  - Flexible 'plug and play' single use technologies
  - Working to implement new technologies into INDs
- Driving to :
  - Small modular agile facilities with embedded sustainability
  - Responsive synchronized value chain for adaptive clinical & commercial production with global reach



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THANK YOU

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